

Reading input in C

Adapted from materials by Dr. Carrier



Where do we get input?

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- stdin
- Command line arguments
- Files

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- **stdin** (our focus here)
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- Files

scanf

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Documentation:

<https://cplusplus.com/reference/cstdio/scanf/>

scanf - example

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```
int x = 0;
int sum = 0;
while(1){
    int res = scanf("%d", &x);
    if(res == EOF) break;
    sum += x;
}
printf("Sum is %d\n", sum);
```

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fgets - example

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```
int main(){
    char s[10];
    while(1){
        char* res = fgets(s, 10, stdin);
        printf("received input: %s\n", s);
        if(res == NULL) break;
    }
}
```

getline

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 - Updates both values

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 - Or -1 for errors

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Documentation:

<https://man7.org/linux/man-pages/man3/getline.3.html>

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void main(){
    while(1){
        char* s = NULL;
        size_t n = 0;
        int res = getline(&s, &n, stdin);
        printf("Received: %s\n", s);
        free(s);
        if(res == -1) break;
    }
}
```

Notes

- Be careful about newlines
 - `scanf(" %c", ...)` is recommended over `scanf("%c\n", ...)`
 - This is why printing strings from `fgets` and `getline` have empty lines after them!
 - If you need to "eat" a character, use [`getchar\(\)`](#)
- Be careful mixing and matching these functions
 - e.g., `fgets` and `getline` consume newlines
 - `scanf` does not
- I prefer `scanf`
 - It makes dealing with non-string types easier
 - We'll talk later about `str->int` conversion